A2 with generative AI: building your SPC workbook in Claude

You may use a generative AI assistant to help build your A2 workbook. This guide shows one way to do it with a free Claude account at claude.ai. It takes about two hours, most of it checking what Claude gives you. Claude does the arithmetic and the Excel formatting quickly. The marks for interpretation, the OCAP and the commentary come from your own judgement about the Johannesburg tableting line, so the guide keeps those parts in your hands.

Before you start

Sign in at claude.ai with a free account. Open Settings, then Capabilities, and turn on **Cloud code execution and file creation**. Claude needs it to read your CSV, run calculations and hand you a real `.xlsx` file. You do not need to add any other skill: Claude's Excel skill switches on by itself when you ask for a workbook.

If Claude stops and asks permission to run a command, click **Allow once**. The command runs in Claude's own sandbox in the cloud, not on your computer. A prompt that builds a file can take 5 to 15 minutes. Leave the tab open until Claude has finished and the file appears in the chat.

Start a new chat for this assignment and keep the whole job in that one chat, so Claude remembers what it has already built. A free account has a usage limit that resets every five hours. If you hit it, wait and carry on in the same chat.

Files to upload

Download these from the A2 assignment page on InnoSkills and attach all four to your first message, using the paperclip button:

1. `A02_spc_control_chart_workbook.pdf`, your brief.
2. `A02_hibiscus_pressure_spc_120.csv`, the 120 tablet measurements.
3. `A02_exemplar_lilongwe_moringa_SPC_Workbook.xlsx`, the worked exemplar workbook.
4. `A02_exemplar_lilongwe_moringa.pdf`, the marker's note on the exemplar.

The exemplar is set at a different plant (a liquid filler in Lilongwe), so its numbers and wording would answer the wrong process. Use it to see the shape of a strong workbook: one job per sheet, every limit a visible formula, signals logged with the rule that fired.

Prompts

Copy each prompt into the chat in turn. Read the answer before you send the next one. Where a prompt has a square-bracketed gap, fill it in yourself.

1. Read the brief

```
I'm a learner on the DigiSkills programme. Attached are my A2 brief (PDF), its dataset (CSV), and a worked exemplar workbook plus its marker's note from a different plant. Read the brief carefully. Then give me a checklist of every Key Requirement and every row of the marking scheme, with what a Distinction answer needs for each. Do not build anything yet. My submission is about the Johannesburg Hibiscus Pressure tableting line, not the exemplar's Lilongwe fill line.
```

2. Check the data

```
Load A02_hibiscus_pressure_spc_120.csv with code. Tell me: the column names, the specification limits stated in the file header, how many subgroups and how many measurements per subgroup, and any missing values, duplicates or values that look like transcription errors. Do not clean anything yet; just report what you find and suggest how each problem should be handled.
```

Decide for yourself how to handle each problem Claude reports, and tell it in the next prompt. That decision is part of your mark.

3. Build the workbook

```
Build an Excel workbook named A02_SPC_Workbook_[YourSurname].xlsx for hardness and weight. Sheets, in order: README (what each sheet does and how the data was checked), Data (the CSV untouched), Clean (my cleaning decisions: [describe them]), Limits (a table of the A2, D3, D4 and d2 constants looked up for n, and the grand mean, R-bar and control limits as live formulas), X-bar and R charts for each characteristic with centre line, UCL and LCL plotted from the limit columns and out-of-control points marked, Rules (at least four Western Electric/Nelson rules as helper columns, one column per rule), Signal_Log (each signalled subgroup with the rule that fired), Capability (Cp, Cpk, Pp and Ppk as formulas against the spec limits), OCAP, DoE and Commentary (leave OCAP, DoE and Commentary as empty headed sheets for now). Every calculated cell must be an Excel formula, not a pasted number. Follow the exemplar's layout ideas but none of its numbers or wording.
```

4. Check the numbers independently

```
Now recompute the grand mean, R-bar, the X-bar and R control limits, and Cp, Cpk, Pp and Ppk for both characteristics directly from the CSV in Python, without reading the workbook. Then open the workbook, recalculate its formulas, and show me a table comparing the two. Flag any difference.
```

If anything disagrees, ask Claude to find and fix the cause, then run this prompt again.

5. Design the experiment

```
Help me design the DoE sheet: a replicated 2 x 2 factorial experiment on Hibiscus Pressure tablet hardness. My two factors are [for example compression force and dwell time] at levels [your levels], with [n] runs per cell, randomised. Build the design matrix and a two-way ANOVA block as live formulas that calculate once trial results are typed in. Do not invent trial results. Explain how I should read main effects and the interaction.
```

6. Write the OCAP and commentary yourself, then get feedback

Write the OCAP and the commentary in your own words, using what your charts show. Then paste each draft into the chat:

`Here is my draft [OCAP / commentary]. Mark it against the "[OCAP Design / Commentary and Risk Assessment]" row of the marking scheme in my brief. Tell me which band it reaches and what would lift it one band. Point out any figure in it that does not match the workbook. Do not rewrite it for me. My draft: [paste your draft here]`

When you are happy with a draft, ask Claude to place it on its sheet and format the OCAP to print on one A4 page.

Check before you submit

- Open the file in Excel or LibreOffice Calc. Click a few limit and capability cells and confirm each one shows a formula, not a typed number.
- Work out one control limit by hand from R-bar and the A2 constant, and check it matches.
- Look at every chart. Each needs a centre line, both limits and the out-of-control points marked.
- Read every sheet for leftover exemplar content: Lilongwe, moringa, fill volume or ml should not appear anywhere.
- Print-preview the OCAP sheet and confirm it fits one page.
- Name the file `A02_SPC_Workbook_[YourSurname].xlsx` with your own surname.

Add a short note on the README sheet saying which parts Claude built and which decisions and writing are yours. Claude can make mistakes. You are responsible for every number and every sentence in what you submit.